



Removing Delivery Bottlenecks

Week 7 • Day 5 • DP ships today

TPM Academy



Day 5 Objectives

- Convert top 3 queues into **3 experiments** with hypothesis / test / success criterion
- Apply the **Theory of Constraints**: one bottleneck at a time
- Run the **integration pass** across Delivery Plan (DP) Sections 1–5
- Cross-review + sign off the DP — the 5th sibling artifact
- Declare the **Week-8 capstone candidate**



Today's Shape

Clock	Block	Outcome
09:00 – 09:15	Opening	Pin DP Sections 1–4
09:15 – 10:30	Block 1 + Activity 1	Design 3 experiments
10:45 – 12:00	Block 2 + Activity 2	DP integration pass
13:00 – 14:15	Block 3 + Activity 3	Cross-review with rubric
14:30 – 16:00	Block 4 + Activity 4 + Wrap	Sign-off + Week-8 candidate

Block 1 — Theory of Constraints

One bottleneck at a time



Goldratt's Claim

A system's throughput is constrained by **exactly one bottleneck at a time**. Not many. One. Optimize anything else and you don't speed up the system — you add inventory in front of the constraint.

Software example: if your slowest step is "code review takes 2 days," speeding up coding doesn't help — it creates more code waiting for review.

Find the bottleneck. Move it. Then find the next one.



Experiment Structure

```
### Experiment N: <name>
**Bottleneck addressed:** <queue>

<p><strong>Hypothesis:</strong> We believe that <x> will <result>
because <reasoning>.</reasoning></result></x></p>
<p><strong>Test:</strong> <what / how long / what scope></p>
<p><strong>Success criterion:</strong></p>
<ul>
<li>Baseline: <current></current></li>
<li>Target: <improved></improved></li>
<li>Window: <duration></duration></li>
</ul>
<p><strong>What could go wrong:</strong> 2-3 risks
<strong>Who runs it:</strong> named person
<strong>By when:</strong> date</p>
<p><strong>Decision after:</strong></p>
</queue></name>

<li>Met: adopt / scale / institutionalize</li>

• Not met: abandon / iterate / escalate
```

"We'll see if it feels better" is not an experiment.

Activity 1 — Design 3 Experiments

Triads • 35 minutes

- 5 min: re-read top 3 queues
- 25 min: draft 3 experiments (template)
- 5 min: would your team actually try these in 60 days?



5 + 25 + 5

Block 2 — DP Integration Pass

Sections 1–5 reads coherently top to bottom



The Integration Checklist

The integration pass verifies Sections 1–5 cohere, including the value stream mapping (VSM).

Check	Pass criterion
Section 1 outcomes ↔ Section 2 backlog	Every output in Section 2 maps to an outcome in Section 1
Section 2 backlog ↔ Section 3 tracking	Every leading indicator corresponds to backlog
Section 3 tracking ↔ Section 4 VSM	Cycle-time leading indicators consistent with VSM
Section 4 queues ↔ Section 5 experiments	Every top queue has an experiment
References to PRD/TCD/TMD/SEP	DP cites prior artifacts by section
No fortune-cookie prose	Specific to this feature

Activity 2 — Integration Pass

Triads • 40 minutes

- 15 min: solo read-through
- 10 min: pool issues
- 15 min: fix in priority order (coherence first)



15 + 10 + 15

Block 3 — Cross-Review

The Friday DP Rubric



The DP Rubric

Dimension	Weight
Outcome clarity	20%
ADO discipline	15%
Output ↔ outcome pairing	20%
Value stream realism	20%
Bottleneck experiments	15%
Integration with prior artifacts	10%

Score thresholds: ≥ 3.0 ships. 2.0–2.9 with named gaps. $< 2.0 \rightarrow$ facilitator review.

Activity 3 — Cross-Review

Triad pairs • 40 minutes

- 5 min: pair set-up
- 20 min: cross-read with rubric
- 10 min: author triad responds
- 5 min: capture final score



5 + 20 + 10 + 5

Block 4 — Sign-Off + Week-8 Bridge

The 5-artifact set ships; capstone candidate declared



What Carries to Week 8

Carries forward

- Artifact templates — Product Requirements Document (PRD), Technical Concept Document (TCD), Technical Modeling Document (TMD), Stakeholder Engagement Plan (SEP), DP — in shorter form
- AI provenance discipline
- Negotiation / outcome / tracking muscles
- Cumulative AI-validation log

New in Week 8

- **AI Spec Development** — using AI to draft technical specs that integrate the artifact set
- A capstone subject (FieldPulse or your own project)
- Final artifact presentations

Pick Your Capstone Candidate

Each triad picks 3–5 candidates and declares a preferred one for Monday.

- **Real product / feature** from work, school, hobby, public
- **Substantive enough** for a week of work
- **Clear problem** you can state in one sentence
- **Public-discussable** (no non-disclosure agreement (NDA) conflicts)

Activity 4 — Sign-Off + Capstone Candidate

Triads • 45 minutes • Wrap

- 10 min: adopt / defer / reject reviewer findings; sign off
- 15 min: brainstorm 3–5 capstone candidates
- 10 min: pre-flight (inputs needed; people to ask)
- 5 min: declare preferred candidate



10 + 15 + 10 + 5 • 15-min wrap



Week 7 Wrap

What we shipped this week

- Outcome map (DP Section 1)
- Loaded Azure DevOps (ADO) backlog (Section 2)
- Tracking plan (Section 3)
- Value stream map (Section 4)
- Bottleneck experiments (Section 5)
- Integrated DP — 5th sibling artifact

What opens Week 8

- **AI Spec Development & Capstone**
- Capstone subject (FieldPulse or your own)
- AI Spec dev integrated with artifact set
- Final artifact presentations Friday

Bring to Monday: All 5 sibling artifacts (PRD/TCD/TMD/SEP/DP) + your declared capstone candidate.

Questions & Reflection

Which bottleneck have you been living with that you no longer accept?